

How Quantum Computing Threatens Today's Cryptography and What It Means for Crypto Investors

Introduction: A New Kind of Threat

Quantum computing (QC) breakthroughs in 2025 have captivated both technologists and investors, offering incredible promise but also igniting fresh anxieties. One growing concern centers on cryptography: the mathematical foundation that secures everything from online banking to cryptocurrencies like Bitcoin and Ethereum.

As quantum computers edge closer to real-world viability, investors are asking: *Could tomorrow's machines undermine today's blockchains?*

Quantum's Superpower: Breaking Encryption

Quantum computing allows for a fundamentally different approach to computation. Unlike classical computers, which process bits one at a time, quantum systems use [qubits](#) and leverage the laws of [superposition](#) and [interference](#) to evaluate countless possibilities and drown out incorrect figures, simultaneously. This enables them to perform certain tasks exponentially faster than classical computers.

The key vulnerability of today's security infrastructure is [public key cryptography](#) —specifically algorithms like RSA (prime-factor based) and ECDSA (elliptic-curve based)— that secure both traditional financial applications and blockchains.

To visualize it, imagine a massive padlock whose combination is the product of two giant prime numbers. A classical computer would take billions to trillions of years to crack it.¹ But with [Shor's algorithm](#), a quantum computer could factor those primes to 'break the lock' in hours, or even minutes under the most optimistic hardware assumptions.

In other cases, quantum computers don't break the lock entirely—but they do make it easier to pick. Grover's algorithm, for instance, doesn't fully crack hash functions like SHA-256, but it reduces their effective strength by half—making brute-force attacks significantly more feasible over time.

By mid-2025 the cutting edge of gate-model hardware is still counted in mere thousands of **physical** qubits: IBM's 1,121-qubit *Condor* chip is in the lab and its 1,386-qubit multi-chip *Kookaburra* processor is scheduled to debut later this year, while Google's 105-qubit *Willow* device has demonstrated a single error-corrected logical qubit that can persist for about an hour—an impressive milestone but still only one logical qubit. [IBM's Technica](#) Even with aggressive low-density parity-check codes, these machines translate to well under ten logical qubits today, and IBM's own roadmap targets roughly **200 logical qubits** by year-end 2025. [The Quantum Insider](#) To put that in perspective, factoring an RSA-2048 modulus with Shor's algorithm still needs

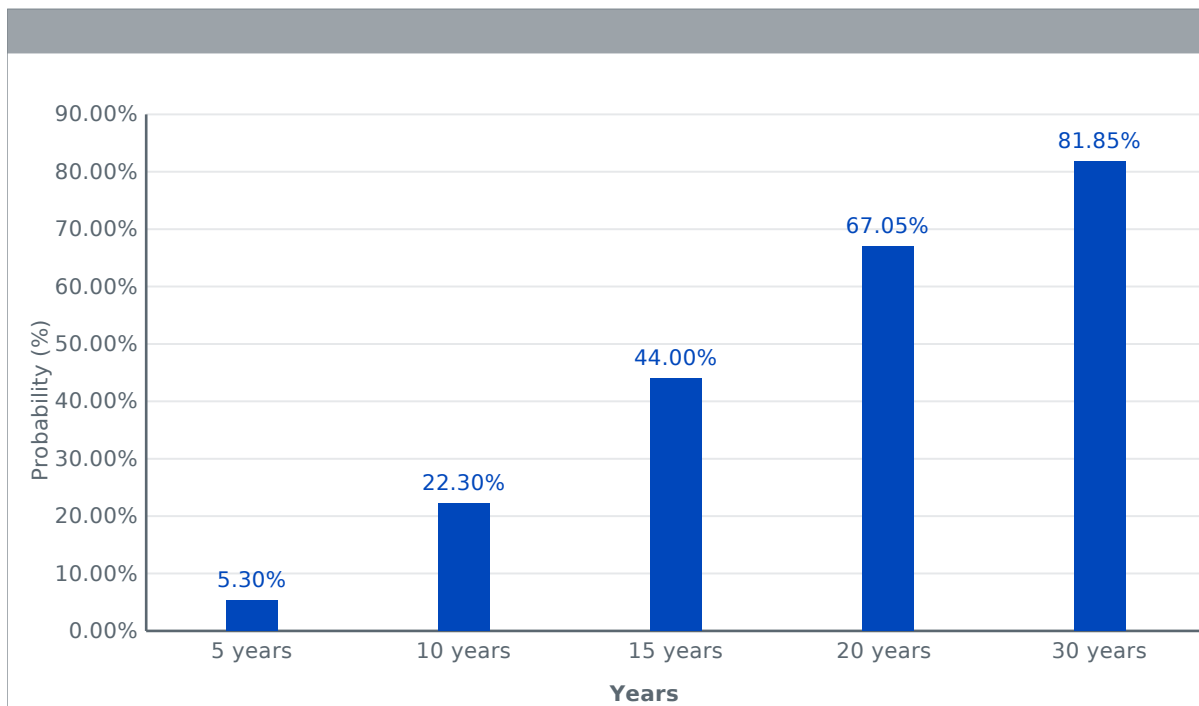
¹ QuintessenceLabs, Breaking RSA Encryption – an Update on the State-of-the-Art, QuintessenceLabs Blog, June 13 2019.

on the order of **1,700–3,000 logical qubits** ($\approx 10^8$ fault-tolerant gates), which—after overhead—maps to $\lesssim 1$ **million physical qubits** under the most optimistic May-2025 resource estimate (down from the 20 million widely quoted just a few years ago). [arXivPostQuantum.com](https://arxiv.org/postquantum) Breaking the elliptic-curve signatures behind Bitcoin and TLS (ECDSA-256) is “easier” but would still require **~1,500–4,000 logical qubits**, translating to hundreds of thousands of high-fidelity physical qubits—two orders of magnitude beyond any system we can build today. [CryptoTickerscaleap.io](https://cryptotickerscaleap.io) In short, today’s prototypes have finally crossed the error-correction threshold, yet they remain thousands of logical qubits short of threatening RSA or ECDSA, giving the world a meaningful—but not indefinite—window to deploy post-quantum cryptography.

While the math is proven, the necessary hardware does not yet exist. However, the accelerating pace of quantum computing progress poses a growing risk to today’s encryption standards.

How Soon Is the Threat?

The benchmark for a “serious quantum threat” is a quantum machine capable of breaking RSA-2048 encryption, the internet’s standard encryption, within 24 hours. Consensus expert forecasts show the chances for a serious quantum threat in the next 5, 10, 15, and 20 years: 5%, 22%, 44%, and >67%, respectively.²



Source: *Blockchain Security Risk Assessment in Quantum Era, Migration Strategies and Proactive Defense* - Yaser Baseri, Abdelhakim Hafid, Yahya Shahsavari, Dimitrios Makrakis and Hassan Khodaiemehr

² Yaser Baseri, Abdelhakim Hafid, Yahya Shahsavari, Dimitrios Makrakis and Hassan Khodaiemehr, *Blockchain Security Risk Assessment in Quantum Era, Migration Strategies and Proactive Defense*, Figure 4 “Expected Likelihood of Quantum Threat for Classic Cryptography,” p. 6–7.

If RSA were broken by a large-scale quantum computer, the digital signatures and key exchanges underpinning banking, secure e-mail, VPNs, and many authentication systems could be forged or decrypted, triggering systemic cyber risk. The same capability could also break [ECDSA](#) and other elliptic-curve signatures used by Bitcoin and Ethereum.

However, not all systems are exposed on equivalent timelines. Cracking a Bitcoin key fast enough to front-run a pending transaction—within its ~10-minute block time—requires significantly more quantum power than decrypting stored RSA keys over several days. Financial infrastructure likely falls somewhere in between, with many systems operating on intraday cycles and rotating keys regularly.

What Happens to Bitcoin and Ethereum in a Post-Quantum World?

If ECDSA is broken, blockchain integrity collapses. Attackers could forge digital signatures to authorize fake transactions, as well as manipulate old blocks, effectively rewriting transaction history.

Security experts often frame these risks using the STRIDE model, which outlines key categories of threats:

- **Spoofing** – when an attacker impersonates a user’s private key
- **Tampering** – unauthorized modification of the blockchain ledger
- **Repudiation** – denial of responsibility for forged or fraudulent transactions
- **Denial-of-Service (DoS)** – disruption of mining operations or consensus mechanisms
- **Elevation of Privilege** – unauthorized control or takeover of network nodes

Threats Posed to Bitcoin and Ethereum in STRIDE Terms

Platform	Vulnerable Components	Potential Impact	STRIDE Threats	L	I	R
Bitcoin	ECDSA: Used for digital signatures SHA-256: Used for hashing blocks	Forgery of digital signatures, enabling unauthorized spending Breaking collision resistance of SHA-256, disrupting mining and block verification	Spoofing: Impersonation through compromised signatures			
			Tampering: alteration of transaction data or blocks			
			Repudiation: Disputes arising from forged transactions	H	H	H
			DoS: Mining and consensus disruption			
			Elevation of Privilege: Unauthorized access to smart contracts			

Ethereum	ECDSA: Used for digital signatures	Signature forgery and potential disruption of consensus mechanism	Spoofing Identity impersonation via ECDSA compromise			
	Keccak-256: Used for hashing transactions and blocks	Breach of encrypted communication	Tampering: Block manipulation by breaking Keccak-256			
	Elliptic Curve Integrated Encryption Scheme (ECIES): Encrypts communication (less common)	Tampering with smart contract states or inputs	Repudiation: Disputed from forged or altered transactions			
			Information Disclosure: Leaked data through broken ECIES encryption	H	H	H
			DoS: Quantum attacks degrading network operations			
			Elevation of Privilege: Unauthorized access to smart contracts			

Source: *Blockchain Security Risk Assessment in Quantum Era, Migration Strategies and Proactive Defense* - Yaser Baseri, Abdelhakim Hafid, Yahya Shahsavari, Dimitrios Makrakis and Hassan Khodaiemehr. **Key: L = Likelihood, I = Impact, and R = Risk.**

Taken together, these breaches would break trust in the blockchain, and the ensuing fallout could trigger cascading liquidity shocks across the ecosystem. However, because breaking encryption in a live network requires more advanced quantum systems capable of real-time attacks, blockchains may have more time to implement quantum defenses before the threat is realized.

The Defense: Post-Quantum Cryptography (PQC)

Quantum threats may raise concern amongst investors, but cryptographers are fighting back with post-quantum cryptography. PQC, broadly speaking, is any algorithm based on a math problem that would be too difficult for both classical and quantum computers to solve. NIST³ has already selected four main algorithms for PQC to be initially standardized; three are based on [structured lattices](#), and the fourth is hash-based.⁴ These are designed to replace vulnerable existing algorithms like RSA and ECDSA. Implementing these quantum-resistant methods would significantly reduce the risks posed by quantum computing and help secure both blockchain networks and traditional financial security infrastructure.

³ NIST – The National Institute of Standards and Technology

⁴ National Institute of Standards and Technology, *NIST Post-Quantum Cryptographic Candidates to Be Standardized: Round 4* (2022). See discussion of Kyber, Dilithium, Falcon and SPHINCS+ in Section B, “Selected Quantum-Resistant Cryptographic Standards,” Table VI, p. 8 of *Quantum Blockchain Security Research*

Upgrading Blockchains: The Governance Hurdle

The core difference between banks and blockchains isn't cryptographic vulnerability, it's how they respond to it. While banks can coordinate top-down upgrades across their infrastructure, blockchains operate through decentralized consensus, so the lead time on a shift of this magnitude could take years.⁵ Blockchain rule changes must win the support of thousands of independent participants. This distributed governance model, while a strength in terms of resilience and transparency, risks slowing transitions like PQC adoption.

Adding to the complexity, on-chain **PQC would only shield future activity from attacks** and could not retrofit legacy encryption. Roughly 25% of Bitcoin's supply is already exposed through known public keys, meaning those users must manually transfer their coins to quantum-safe addresses.⁶ This adds yet another layer to the adoption challenge. One proposed solution is an "upgrade-or-freeze" model, which would lock untransferred coins after a grace period, but implementing such a policy could be politically difficult.

Still, progress is being made: last month, Chris Papathanasiou, founder of [BitcoinQS](#), and other cryptography experts released [BIP-360](#), a proposed three-phase, five-year migration plan for Bitcoin. First, it would prohibit new transfers to legacy addresses not protected by PQC. After the five-year grace period, any coins still sitting in vulnerable wallets would be locked forever, rendering that currency unspendable. But, a potential future phase currently being studied might let owners reclaim those frozen coins. With proposals like BIP-360 gaining traction, the path toward quantum resilience is materializing.

Implementation of BIP-360

Phase	What Happens	Who Must Act	Time Horizon
A	Permitted sends are from legacy scripts to P2QRH scripts.	Everyone holding or accepting BTC.	3 years after BIP-360 implementation.
B	At a predetermined block height, nodes reject transactions that rely on ACDSA/Schnorr keys.	Everyone holding or accepting BTC.	2 years after Phase A activation.
C	Users with frozen quantum vulnerable funds and a HD wallet seed phrase can construct a quantum safe proof to recover funds.	Users who failed to migrate funds before Phase B.	TBD pending research, demand, and consensus.

Source: *A Post-Quantum Migration Proposal* via Jameson Lopp

Key:

P2QRH - Pay-to-Quantum-Resistant-Hash

Schnorr - Schnorr signature algorithm (BIP-340)

⁵ Amin Haqshanas, "Bitcoin BIP proposes quantum-resistant upgrade by 2030," Cointelegraph, July 16 2025.

⁶ Deloitte, Quantum computers and the Bitcoin blockchain, Deloitte Risk Advisory (Netherlands), authors Itan Barmes, Bram Bosch & Olaf Haalstra, 2021.

HD - Hierarchical Determinant (wallet)

Traditional Finance: Not Immune, but Moving Forward with Centralized Control

Traditional finance is no safer than blockchains. Both still ride on RSA / ECC, but the centralized command structure of traditional institutions creates the opportunity to mobilize faster. The Bank for International Settlements recommends that the entire financial sector move to PQC now.⁷ Despite the guidance, a recent [survey](#) shows 87% of enterprises – including banks – deem adopting PQC urgent, yet less than half are working towards it today.

Some leaders are moving: JPMorgan is piloting a [quantum-secured network](#) that combines post-quantum algorithms with quantum-key distribution, while Mastercard is building similar protections into its payments infrastructure. Others are likely to follow as auditors like KPMG begin flagging “[harvest-now, decrypt-later](#)” risk as a major liability.⁸ But so far, the ability to move fast is only beginning to translate into action for much of the industry.

Conclusion: Real Risk, But Real Progress

Both blockchains and traditional finance face serious cryptographic vulnerabilities as quantum computing advances. Internet-scale RSA could fall first, giving centralized institutions a narrow window to adapt. Live blockchain exploits are likely further off, giving blockchains more time to implement quantum defenses—even if decentralized governance risks slowing coordination.

PQC offers a viable path forward, and adoption is picking up pace—with the market projected to grow nearly tenfold by 2030. For blockchains, decentralized governance may slow implementation, but proposals like BIP-360 show coordinated upgrades are possible. Traditional institutions may be positioned to move faster, though many have yet to progress in a meaningful way.

It's easy to dismiss the threat of quantum computing as something distant and abstract—something for cryptographers, not CIOs. But look at the capital flows and partnerships emerging today, and you'll see the opposite story playing out. SandboxAQ is embedding quantum-safe scanning and migration tools across the enterprise stack via global consulting giants. IBM is hardwiring quantum-resilient tooling into mainframes and OpenShift containers. PQShield has moved from academic papers to secure boot IPs inside Arm chips. QuSecure is protecting U.S. defense networks with post-quantum VPNs. Quantinuum is injecting quantum randomness directly into production signing keys. These are not theoretical moves. They're infrastructure-grade deployments—critical systems being hardened today against a computational model that doesn't yet exist, but absolutely *will*.

This matters because encryption isn't like semiconductors—you can't upgrade it with the click of a patch. Migrating to public key infrastructure takes years, especially in regulated industries. The systems doing it first—telecoms, aerospace, banking—aren't paranoid; they're planning around time asymmetry. Quantum advantage might not crack RSA or ECDSA in 2025 or even 2030, but the migration clock starts now. If you're allocating capital with a three-to-five-year horizon and ignoring this transition, you're behind the curve. The lesson here isn't fear of quantum, it's a recognition of entropy: the cost of inaction increases every day the infrastructure doesn't evolve. Watch where the most forward-leaning cryptographic infrastructure is being built—because that's where resilience and opportunity are converging.

⁷ Bank for International Settlements, “Quantum-readiness for the financial system: a roadmap,” BIS Papers No. 158 (July 2025), 3 – 4 (“a CRQC may be realised as soon as in the next decade... organizations must urgently initiate preparations today”)

⁸ KPMG, *Emerging Tech Risk – Quantum*, KPMG Services Pte. Ltd., October 2024, p. 4.

As the quantum era approaches, investors may want to barbell their exposure—maintaining ownership of both their crypto assets and the companies building quantum-safe infrastructure—to manage risk and participate in the transformation ahead.